

Part I

## Recitation Questions

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### Problem 1

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February 21, 2026

*Problem 1*

**Problem 1** *Compute the following indefinite integrals.*

(a)  $\int 2t \sin(t^2) \, dt.$

(b)  $\int \cos(x) \sqrt{\sin(x)} \, dx.$

(c)  $\int \left( 3t^2 - 4 + \frac{1}{t} \right) e^{t^3 - 4t + \ln(t) - 9} \, dt.$

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## Problem 2

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February 21, 2026

*Problem 2*

**Problem 2** Evaluate the following definite integrals:

(a)  $\int_{-2}^1 t^2 \sin(t^3) \, dt$

(b)  $\int_0^{1/2} \frac{13e^u}{3e^u - 5} \, du$

(c)  $\int_0^{\pi/2} \frac{\cos(x)}{4 + \sin(x)} \, dx$

(d)  $\int_1^4 \frac{e^{\sqrt{x}}}{3\sqrt{x}} \, dx$

(e)  $\int_{\pi/3}^{\pi/2} \sin(x) \sec^2(\cos(x)) \, dx$

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# Problem 3

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February 21, 2026

Problem 3

**Problem 3** Suppose that  $\int_1^3 f(x) dx = 4$

(a) Evaluate the following integrals.

(i)  $\int_0^2 f(x+1) dx$

(ii)  $\int_1^9 3 \frac{f(\sqrt{x})}{\sqrt{x}} dx$

(iii)  $\int_{\frac{1}{3}}^1 f(3x) dx$

(iv)  $\int_2^4 3f(x-1) dx$

(v)  $\int_0^{\sqrt{2}} 3xf(x^2+1) dx$

(b) Assume that  $f$  is odd. Evaluate  $\int_{-3}^{-1} f(x) dx$

(c) Assume that  $f$  is even. Evaluate  $\int_{-3}^{-1} f(x) dx$

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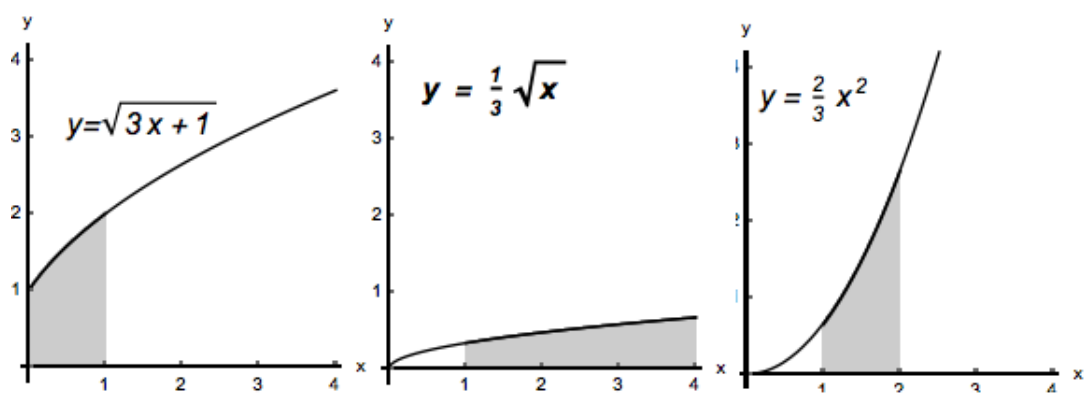
# Problem 4

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February 21, 2026

Problem 4

**Problem 4** Show that the areas of the three shaded regions in the figure below are equal. Find this area.





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# Problem 5

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February 21, 2026

Problem 5

**Problem 5** Find the error in the following “solution”:

Find  $\int_{-2}^2 \frac{1}{x^8 - 1} dx$ .

let  $u = x^4$   
then  $du = 4x^3 dx$

$\frac{du}{4x^3} = dx$

and  $u^{1/4} = x$

$u(2) = 2^4 = 16$

$u(-2) = (-2)^4 = 16$

$= \int_{16}^{16} \frac{1}{(u^2 - 1)} \cdot \frac{du}{4u^{3/4}}$

but an integral  $\int_a^a f(x) dx = 0$

so  $\int_{-2}^2 \frac{1}{x^8 - 1} dx = 0$

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# Problem 6

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February 21, 2026

Problem 6

**Problem 6** Find the error in the following “solution”, correct the error (or errors) and evaluate the integral.

Find  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x) \sqrt{1 - \cos^2(x)} dx$ .

“Solution”:

Since  $\sin(x) = \sqrt{1 - \cos^2(x)}$ , let  $u = \sin(x)$ . Then  $du = \cos(x)dx$ ;

when  $x = -\frac{\pi}{2}$ ,  $u = \sin\left(-\frac{\pi}{2}\right) = -1$ ; when  $x = \frac{\pi}{2}$ ,  $u = \sin\left(\frac{\pi}{2}\right) = 1$ . Therefore

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x) \sqrt{1 - \cos^2(x)} dx = \int_{-1}^1 u \cdot du = 0,$$

due to symmetry (integrating an odd function over a symmetric interval,  $[-1, 1]$ ).

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# Problem 7

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*Problem 7*

**Problem 7** Compute the integral:  $\int \frac{1+3x}{4+4x^2} dx$ .

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# Problem 8

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February 21, 2026

*Problem 8*

**Problem 8** Evaluate the integral  $\int \frac{x^2}{1+x^2} dx$ .



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# Problem 9

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February 21, 2026

*Problem 9*

**Problem 9** Compute the integral:  $\int \sec^2(x) \tan(x) \, dx$ .

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# Problem 10

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February 21, 2026

*Problem 10*

**Problem 10** *What are two substitutions that can be used to evaluate the integral*

$$\int x\sqrt{x+8} \, dx$$

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# Problem 11

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February 21, 2026

*Problem 11*

**Problem 11** Compute the integral:  $\int \frac{x}{\sqrt{x-4}} dx$ .

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# Problem 12

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February 21, 2026

**Problem 12** Evaluate the following indefinite integrals:

(a)  $\int \frac{13x^7}{\sqrt{3x^4 - 5}} dx$

(b)  $\int \frac{u^3}{u^2 - 3} du$

(c)  $\int e^{t^2 + \ln(t)} dt$